

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Explain in detail the different types of measurement errors and their significance in instrumentation.

Q.20 Describe the working principle, circuit and application of a digital multimeter.

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Roll No.

2nd Sem/ DVOC Level-5 (Medical Image Tech.)

Subject : Electronic Measurement & Instrumentation-II

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple choice questions. All questions are compulsory (5x1=5)

Q.1 What is the SI unit of electrical resistance?

- a) Ampere b) Volt
- c) Ohm d) Watt

Q.2 Which of the following is an example of a systematic error?

- a) Human misreading
- b) Instrumental error
- c) Fluctuating temperature
- d) Random noise

Q.3 What type of meter is a PMMC instrument?

- a) Moving coil b) Moving iron
- c) Induction type d) Electrostatic

Q.4 Which of the following devices is used for measuring high-frequency signals?

- a) Galvanometer
- b) Digital frequency meter
- c) Ohmmeter
- d) DC ammeter

Q.5 What is the main advantage of a digital voltmeter over an analog voltmeter?

- a) Higher accuracy
- b) Lower cost
- c) Requires no power source
- d) Susceptible to noise

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 Define absolute error in measurement?

Q.7 What is the purpose of a galvanometer?

Q.8 What are the key differences between accuracy and precision?

Q.9 What is the role of a probe in an electronic instrument?

Q.10 Why is a digital multi meter preferred over an analog meter?

SECTION-C

Note: Very Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

Q.11 Explain the significance of metric prefixes in measurement?

Q.12 What are the different types of errors in electronic measurement?

Q.13 How does a DC voltmeter work?

Q.14 Describe the working principle of a transistor voltmeter.

Q.15 What are the applications of a digital frequency meter?

Q.16 Explain the concept of resolution in measurement instruments.

Q.17 Describe the advantages and disadvantages of electronic measuring instruments.

Q.18 How does an AC electronic voltmeter differ from a DC voltmeter?